SPY & QQQ: Was QQQ's Extra Risk Worth It? (2000 – 2025)

A comparative analysis of return, risk, robustness over time and cumulative withdrawals.

Ignacio Sicilia · October 2026

Abstract:
This paper compares SPY and QQQ between December 31, 1999 and December 31, 2025 to determine whether QQQ compensated an investor for taking on more risk. With dividends reinvested, QQQ achieved a CAGR of 8.30% versus 7.98% for SPY, but with 1.5 times the volatility and a maximum drawdown of −83%. SPY was superior on all three risk-adjusted ratios (Sharpe, Sortino and Calmar). The result depends on the entry date: QQQ achieved a higher CAGR in 92% of the 10-year windows and only lost in those starting between late 1999 and early 2001. In a scenario of annual profit withdrawals, QQQ withdrew more in total, although it started seven years later than SPY.

Keywords: SPY, QQQ, ETF, risk-adjusted return, Sharpe ratio, maximum drawdown, entry timing.

Disclaimer: *This document is for educational purposes only and does not constitute investment advice. Past performance does not guarantee future results.*

1. Introduction

This analysis compares two ETFs that track the S&P 500 (SPY) and Nasdaq-100 (QQQ) indices, respectively, in order to answer one question: did QQQ compensate an investor for taking on more risk than SPY?

To answer it, five complementary questions are posed: How much did each one return and how volatile was it? What was the worst drawdown and how long did it take to recover? Did the extra return compensate for the extra risk? Does the conclusion depend on the entry date? And which ETF generated more income by withdrawing only the gains?

2. Methodology

2.1 Data and assumptions

- Period: close of December 31, 1999 to close of December 31, 2025 (26 years).
- Both ETFs indexed to 100.
- Dividends reinvested (total return).
- Nominal USD, not adjusted for inflation.
- Source: Yahoo Finance via the yfinance Python library.
- Analysis environment: Google Colab.
- Macroeconomic regimes are defined by peaks and troughs of the S&P 500 (SPY).
- Frequency: daily for drawdowns, monthly for returns and volatility, annual for Scenario B.

2.2 Scenarios

- Scenario A (buy & hold): single investment of 100 USD, with no contributions or withdrawals.
- Scenario B (profit withdrawal): single investment of 100 USD; every December 31, everything above 100 is withdrawn. If the position is worth less, nothing is withdrawn or replenished. Withdrawn amounts are held in cash with no return.

2.3 Metrics

- Return: cumulative return and CAGR.
- Risk: annualized volatility, downside deviation, maximum drawdown and loss from peak at the end of the period.
- Risk-adjusted: Sharpe (Sharpe, 1966), Sortino (Sortino and Price, 1994) and Calmar (Young, 1991) ratios. Risk-free rate: 3-month T-bill, 1.9% average (Federal Reserve, TB3MS series). The ratios are computed over the full period; Sortino uses a 0% threshold for downside returns.
- For Scenario A, sensitivity to the entry date and 193 rolling 10-year windows are evaluated.

2.4 Hypotheses

- Hyp. 1 – Return (Sc. A): SPY achieves a higher CAGR than QQQ.
- Hyp. 2 – Risk-adjusted return (Sc. A): SPY beats QQQ on at least 2 of 3 ratios (Sharpe, Sortino, Calmar).
- Hyp. 3 – Entry timing (Sc. A): QQQ beats SPY in fewer than 50% of the 10-year windows.
- Hyp. 4 – Profit withdrawal (Sc. B): QQQ generates more cumulative withdrawals than SPY.

3. Results

3.1 Return

An investment of 100 USD made on December 31, 1999, with dividends reinvested, reached 737 USD in SPY and 796 USD in QQQ by December 31, 2025. Figure 1 shows that the paths were very different: QQQ stayed below SPY from 2001 to 2020 and only pulled ahead consistently from 2024 onward.

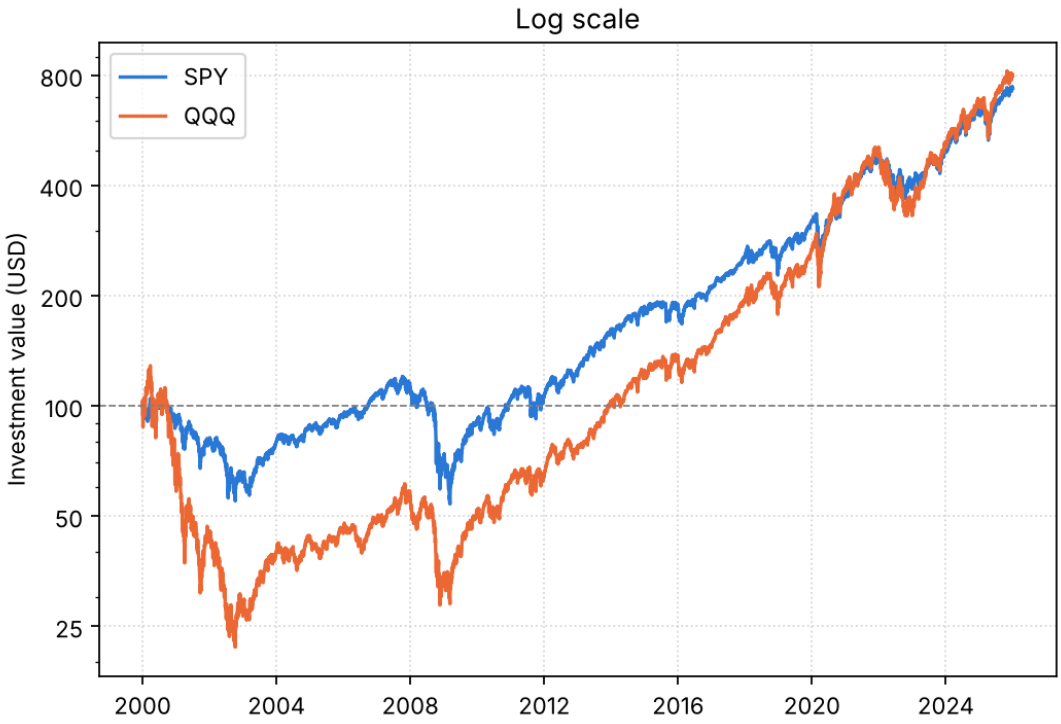


Figure 1. Growth of 100 USD invested on Dec 31, 1999 in SPY (blue) and QQQ (orange), with dividends reinvested. Log scale: the same percentage change takes up the same height anywhere on the chart. The dashed line marks the initial 100 USD. Source: author's calculations based on Yahoo Finance data.

In annual terms, the CAGR was 7.98% for SPY and 8.30% for QQQ (see Table 1). That difference of 0.32 percentage points per year, compounded over 26 years, explains the 59 USD difference in final value (796 vs. 737 USD).

Table 1. Key metrics, Dec 31, 1999 – Dec 31, 2025.

Metric	SPY	QQQ
Final value of 100 USD	737	796
Cumulative return	637%	696%
CAGR	7.98%	8.30%
Annual volatility	15.1%	22.9%
Annual downside deviation	10.2%	15.8%
Maximum drawdown	−55.2%	−83.0%
Years to recover the peak	4.9	14.9
Loss from peak (Dec 31, 2025)	−1.2%	−3.3%
Sharpe	0.46	0.38
Sortino	0.69	0.56
Calmar	0.14	0.10

Source: author's calculations based on Yahoo Finance data.

3.2 Risk

QQQ was more volatile than SPY: its annual volatility was 22.9% versus 15.1%, about 1.5 times higher. Downside deviation, which only considers negative months, shows the same relationship: 15.8% versus 10.2% (see Table 1).

Figure 2 shows how far each ETF fell from its most recent peak. SPY's worst drawdown was −55.2%, between October 9, 2007 and March 9, 2009, and it returned to its peak 4.9 years later. QQQ's was −83.0%, between March 27, 2000 and October 9, 2002, and it took 14.9 years to return to its peak. As of December 31, 2025, SPY was 1.2% below its peak and QQQ 3.3%.

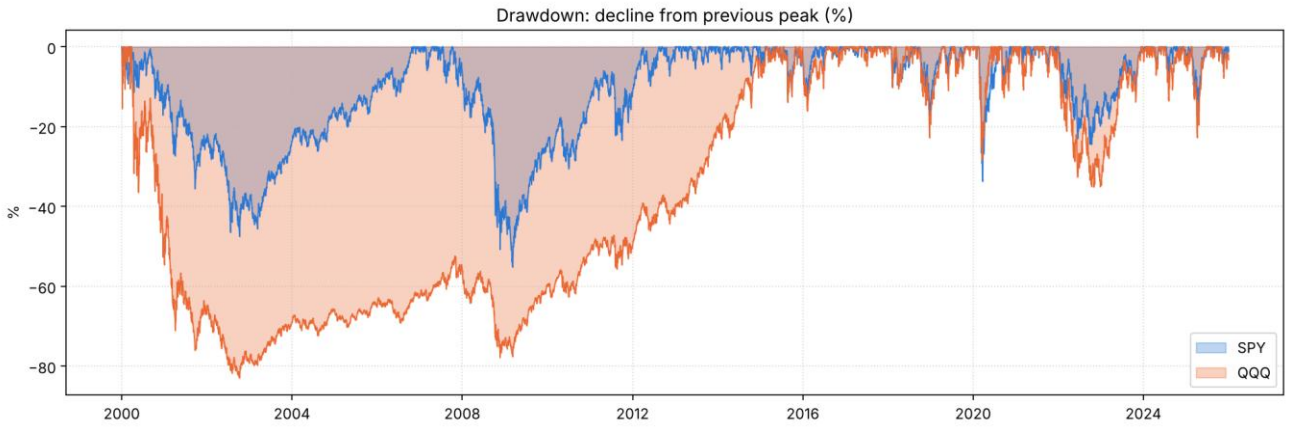


Figure 2. Decline of each ETF from its most recent peak, SPY (blue) and QQQ (orange), 2000–2025. Source: author's calculations based on Yahoo Finance data.

Extreme days were concentrated in crises (Table 2): the six worst days for both ETFs occurred in October 2008 or March 2020, and three of the best in October 2008. October 13, 2008, SPY's best day (+14.5%), came two days before October 15, 2008, one of the worst days for both.

Table 2. The three best and three worst days for each ETF, 2000–2025.

	SPY	QQQ
Best days		
1st	Oct 13, 2008: +14.5%	Jan 3, 2001: +16.8%
2nd	Oct 28, 2008: +11.7%	Jan 7, 2000: +12.4%
3rd	Apr 9, 2025: +10.5%	Oct 13, 2008: +12.2%
Worst days		
1st	Mar 16, 2020: −10.9%	Mar 16, 2020: −12.0%
2nd	Oct 15, 2008: −9.8%	Mar 12, 2020: −9.2%
3rd	Mar 12, 2020: −9.6%	Oct 15, 2008: −9.0%

Source: author's calculations based on Yahoo Finance data.

3.3 Macroeconomic regimes

QQQ outperformed SPY in 6 of the 8 regimes (Figure 3 and Table 3). SPY only did better during the dot-com bubble (−47.5% vs. −82.9%) and the 2022 rate hikes (−24.5% vs. −34.3%).



Figure 3. Growth of 100 USD (top) and decline from most recent peak (bottom), by macroeconomic regime. Red: crisis. Green: expansion. Source: author's calculations based on Yahoo Finance data.

Table 3. Return by macroeconomic regime.

Regime	Period	SPY	QQQ	Type
Dot-com bubble	Mar 2000 – Oct 2002	−47.5%	−82.9%	Crisis
Recovery	Oct 2002 – Oct 2007	+119.0%	+171.0%	Expansion
Financial crisis	Oct 2007 – Mar 2009	−55.2%	−51.6%	Crisis
Low rates and QE	Mar 2009 – Feb 2020	+520.8%	+927.9%	Expansion
COVID	Feb 2020 – Mar 2020	−33.7%	−27.9%	Crisis
Post-COVID stimulus	Mar 2020 – Jan 2022	+119.9%	+137.6%	Expansion
Rate hikes	Jan 2022 – Oct 2022	−24.5%	−34.3%	Crisis
AI boom	Oct 2022 – Dec 2025	+99.9%	+138.7%	Expansion

Source: author's calculations based on Yahoo Finance data.

3.4 Risk-adjusted return

On a risk-adjusted basis, SPY was superior on all three ratios: Sharpe 0.46 vs. 0.38, Sortino 0.69 vs. 0.56 and Calmar 0.14 vs. 0.10 (Table 1). In other words, it earned more return per unit of risk taken (Figure 4).

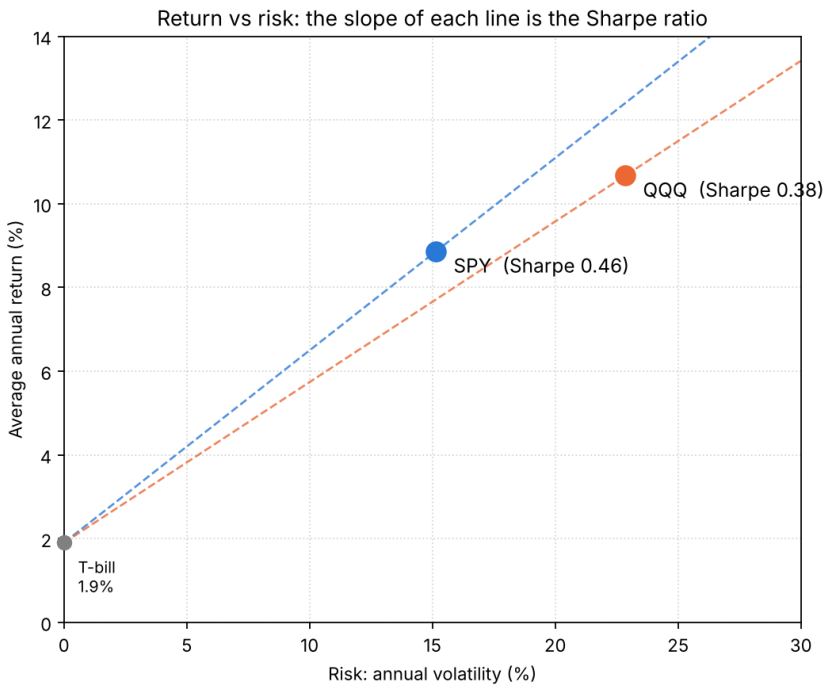


Figure 4. Average annual return versus volatility, 2000–2025. The slope of each line, starting from the T-bill (1.9%), is the Sharpe ratio: the steeper the line, the more return per unit of risk. Source: author's calculations based on Yahoo Finance data.

3.5 Entry timing

For any entry year between 2000 and 2021, holding through 2025, QQQ achieved a higher CAGR than SPY, with differences of +0.3 to +5.9 points per year. Across the 193 10-year windows, QQQ had a higher CAGR in 92% of cases and a better Sharpe ratio in 94%. SPY only came out ahead in windows starting between December 1999 and early 2001 (Figure 5).

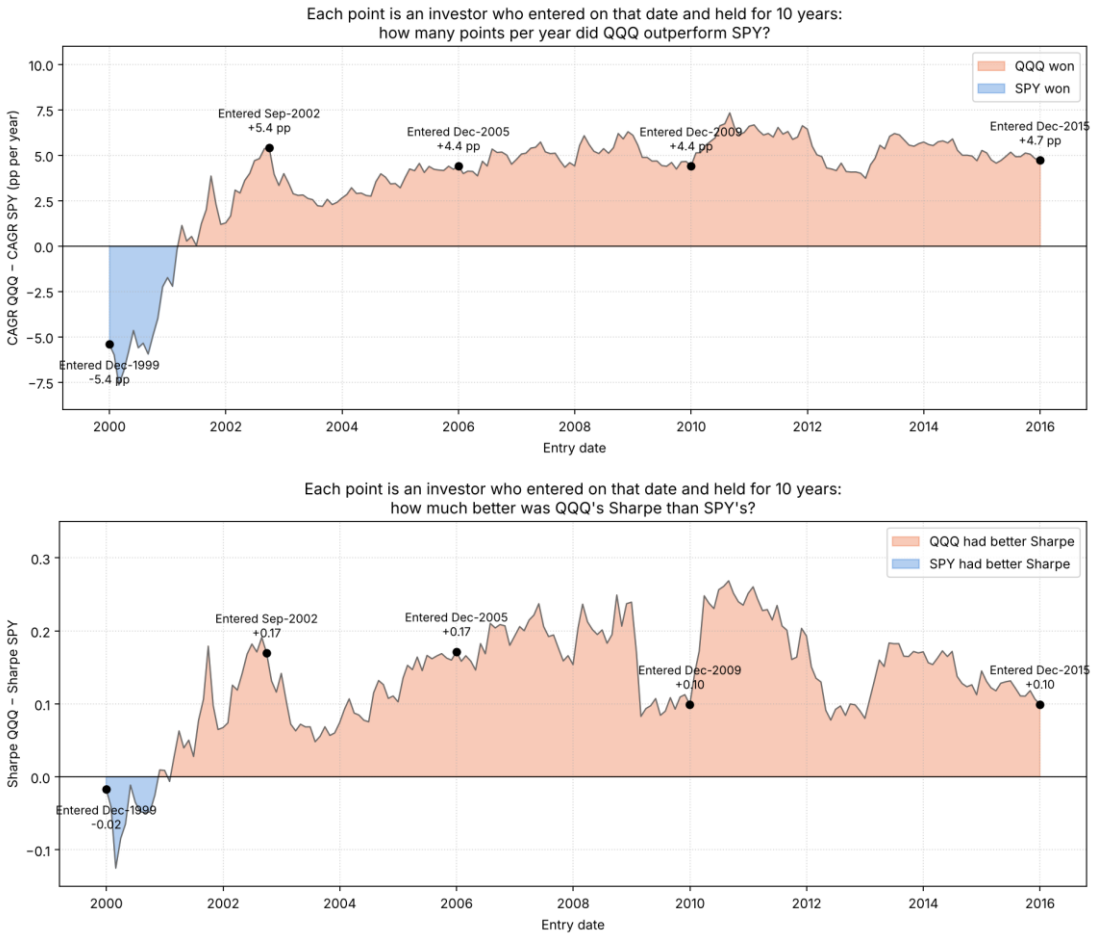


Figure 5. Difference between QQQ and SPY in 10-year investments, by entry date: CAGR (top) and Sharpe (bottom). Each point on the curve is one of the 193 windows; the five marked points are example entry dates. Positive values: QQQ won. Source: author's calculations based on Yahoo Finance data.

3.6 Scenario B

In Scenario B, QQQ accumulated 237 USD in withdrawals versus 220 USD for SPY. SPY made its first withdrawal in 2006 and QQQ not until 2013, because until then its position stayed below the initial 100 USD. QQQ only overtook SPY in cumulative withdrawals in 2020 (Figure 6). For any entry year between 2000 and 2016, QQQ withdrew more than SPY: 8% more when entering in 2000 and between 25% and 50% more for any later entry.

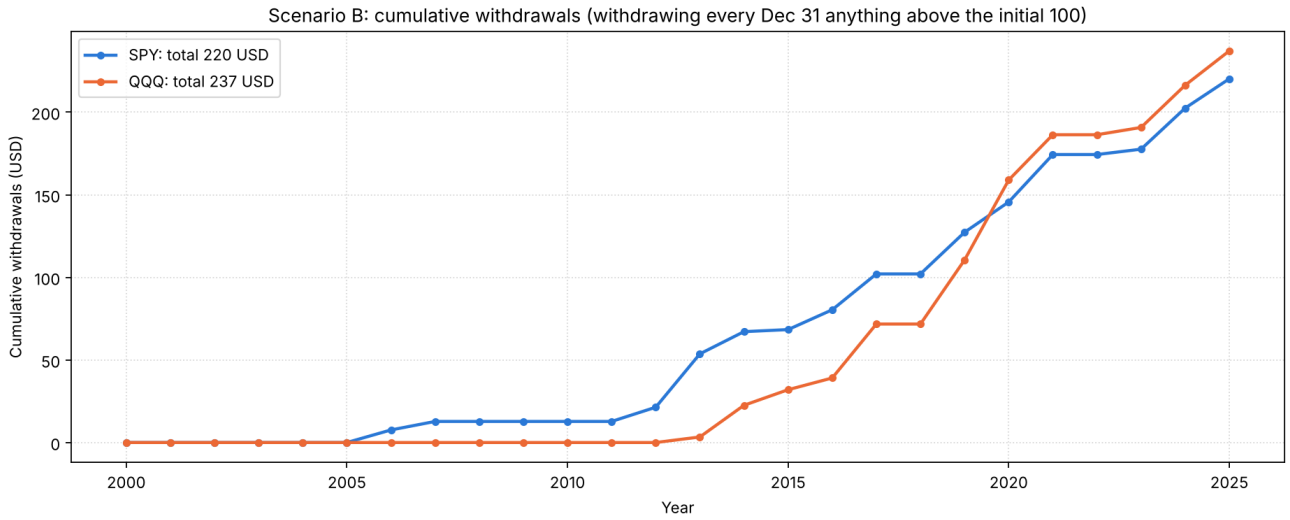


Figure 6. Cumulative withdrawals in Scenario B, SPY (blue) and QQQ (orange), 2000–2025. Source: author's calculations based on Yahoo Finance data.

4. Discussion and conclusion

The answer:

Over the 26 years analyzed, QQQ returned 0.32 percentage points more per year than SPY, a small difference. In exchange, it was 1.5 times more volatile, fell as much as −83% and took almost 15 years to return to its peak. Per unit of risk, SPY performed better on all three ratios. For an investor who entered at the end of 1999, QQQ's extra risk was not compensated.

Why it depends on the starting point:

This result depends on the start date. The period begins just before the peak of the dot-com bubble, the crisis in which QQQ suffered most relative to SPY (−83% vs. −48%) due to its concentration in technology. With any later entry, QQQ achieved a higher CAGR and, in most 10-year windows, a better Sharpe ratio as well. QQQ is neither better nor worse than SPY in general: its outcome is much more sensitive to entry timing.

Crises and extreme days:

The best and worst days were concentrated in the same crises. An investor who sold to avoid the declines probably also missed the strongest rebounds.

Scenario B:

In Scenario B, QQQ withdrew more money in total, but its investor went 13 years without withdrawing anything and only overtook SPY in 2020. SPY delivered income earlier. Table 4 summarizes the outcome of the four hypotheses.

Table 4. Hypothesis outcomes.

Hypothesis	Outcome	Key figure
Hyp. 1 – SPY achieves a higher CAGR than QQQ	Rejected	7.98% vs. 8.30%
Hyp. 2 – SPY beats QQQ on at least 2 of 3 ratios	Confirmed	3 of 3 ratios
Hyp. 3 – QQQ beats SPY in fewer than 50% of the windows	Rejected	QQQ won 92%
Hyp. 4 – QQQ generates more cumulative withdrawals than SPY	Confirmed	237 vs. 220 USD

Source: author's calculations.

In conclusion, QQQ's extra risk translated into a small return advantage over the full period, and its outcome depended far more on entry timing than SPY's.

5. Limitations

- A single historical period is analyzed: the results depend on the start date and do not predict future behavior.
- The composition of both indices changed over 26 years: today's SPY and QQQ do not hold the same companies as in 2000.
- Taxes and trading commissions are not considered.
- Values are in nominal dollars, not adjusted for inflation.
- Scenario B is simplified: withdrawals are annual and withdrawn cash earns no return.
- Macroeconomic regimes are defined by the peaks and troughs of the S&P 500. Measured with QQQ's own peaks and troughs, its drawdowns vary by at most 1.5 points and the conclusions do not change.

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